Answer key · for the teacher

AI basics — answers

AI

Answers for the four pupil worksheets and the unit check. Where pupils write their own ideas, a sample answer shows the kind of thinking to look for — accept any answer that shows the same idea.

Lesson 1 worksheet: AI detective

Spot the AI around you · Learns from examples or follows a rule?

1 Sort it

a. A calculator — **Follows a rule**

b. Voice typing on a phone — **Learns from examples (AI)**

c. A solar lamp that switches on at 6 pm — **Follows a rule**

d. An email spam filter — **Learns from examples (AI)**

e. Keyboard next-word suggestions — **Learns from examples (AI)**

f. A traffic light on a fixed timer — **Follows a rule**

g. An app that spots sick leaves in photos — **Learns from examples (AI)**

h. A school bell that rings at 7:30 — **Follows a rule**

2 Finish the sentence

AI is a computer system that learns **patterns** from **examples** and makes **predictions**.

3 Your turn

Sample answer: A phone camera that finds faces in a photo — it could learn from many photos of faces.

☆ Challenge

Sample answer: It could learn from examples — for example, the times when most pupils actually arrive, or when teachers ring it by hand — and then predict the best time.

Lesson 2 worksheet: Teach the machine

How machines learn from examples · Examples, labels, training and testing

1 Label the examples

a. Football practice moved to 4 pm — **Safe message**

b. You won free airtime. Send your PIN now — **Scam**

c. Your account will close. Click this link — **Scam**

d. Exam timetable is on the notice board — **Safe message**

e. Scholarship approved. Pay a fee to claim it — **Scam**

f. Library books are due back on Friday — **Safe message**

2 Put it in order

1. Collect examples → **2.** Label them → **3.** Train: find patterns → **4.** Test on new examples → **5.** Use: make predictions

3 Finish the sentence

The computer finds **patterns** in the training data and builds a **model**, which we **test** on new examples.

☆ Challenge

Sample answer: Give it many more labelled examples, and more varied ones — different kinds of scams and safe messages. Then train it again and test it on new messages.

Lesson 3 worksheet: Spot the data gap

Good data, fair AI · Whose examples did the machine learn from?

① True or false?

a. Gaps in examples can cause AI mistakes. — **True**
b. A machine can never be unfair. — **False**
c. More of the same examples fixes any gap. — **False**
d. Testing with many groups helps find gaps. — **True**

② What is missing?

a. A fruit app learned only from mangoes and bananas. What is missing? — **Every other fruit — guava, dates, oranges and more. It has never seen them, so it will guess wrong.**

b. Voice typing learned from 1,000 men in one city. Who is missing? — **Women, children, and people from other places with other accents.**

c. A scam filter learned only from English messages. What is missing? — **Scam messages in other languages, such as Kiswahili, Hausa or Arabic.**

③ Match the words

1. Bias → **Works worse for some people**
2. Data gap → **Examples leave some people out**
3. Varied examples → **Every group is in the examples**
4. Testing with groups → **Finding who it gets wrong**

☆ Challenge

Sample answer: Was it tested with pupils like ours — all skin tones and ages, with glasses or headscarves? How often is it wrong for each group, and what happens to a pupil it gets wrong?

Lesson 4 worksheet: Fact detectives

Check before you trust · Sounds sure? Check before you trust.

① True or false?

a. Spiders have six legs. — **False**
b. Water boils at 100 °C at sea level. — **True**
c. Lake Victoria lies entirely in one country. — **False**
d. Mount Everest is the highest mountain in the world. — **True**
e. The Nile flows north into the Mediterranean Sea. — **True**

② Put it in order

1. Ask clearly — no name, phone or photo → **2.** Read the answer and find the facts →
3. Check the facts in a trusted place → **4.** Decide what to use, fix or leave out

③ How would you check?

a. A chatbot says your exam starts on Monday. How would you check? — **Ask my teacher or look at the exam timetable from the school.**

b. A chatbot names a book for your project. How can you tell it's real? — **Look for it in the library or ask the librarian. If nobody can find it, don't use it.**

Challenge

Sample answer: True: “Our town has a market every Saturday.” Made up: “Our town has 500 schools.” My partner spotted the made-up one and said we could check with the head teacher or the local council.

Unit check — answers and reasons

10 questions · the app gives this feedback after each answer

1. **C — Voice typing on a phone.** It learned from many recordings of people speaking.
2. **False.** Automatic is not the same as AI. Many machines only follow fixed steps a person set; most AI learns patterns from examples.
3. **1. Collect examples → 2. Label them → 3. Train: find patterns → 4. Test on new examples → 5. Use: make predictions.** Examples and labels come first; testing on new examples shows whether the model really learned the pattern.
4. **Training data → The examples a system learns from; Label → The answer given to an example, like “spam”; Model → The stored patterns that training produces; Prediction → The system’s best guess — it can be wrong.** These four words describe every machine-learning tool, from spam filters to crop apps.
5. **A — To check it learned the pattern, not just the examples.** New examples show whether it can handle cases nobody showed it.
6. **B — Ask how it was tested, and try it with a few pupils first.** You check whether it works for voices like your pupils’ before anyone is marked by it.
7. **A, B, C. A — yes:** It learns from people like the ones it will serve. **B — yes:** Testing shows who it works worse for. **C — yes:** Reports show where the examples have gaps. **D — no:** Fewer examples usually means more mistakes. **E — no:** That hides the problems other people would meet.
8. **False.** Chatbots predict likely words, so they can sound sure and still be wrong. Check important facts.
9. **check, trusted, decide.** Ask, check, decide — you stay in charge of what reaches your pupils.
10. **A — Pupils’ names, marks and phone numbers.** Personal details about pupils stay out of AI tools.